

BLGE 521 AI

AUTOMATED IMAGE LABELING FOR PRECISION AGRICULTURE

Scaling Crop Phenology Analysis via Hybrid Architectures

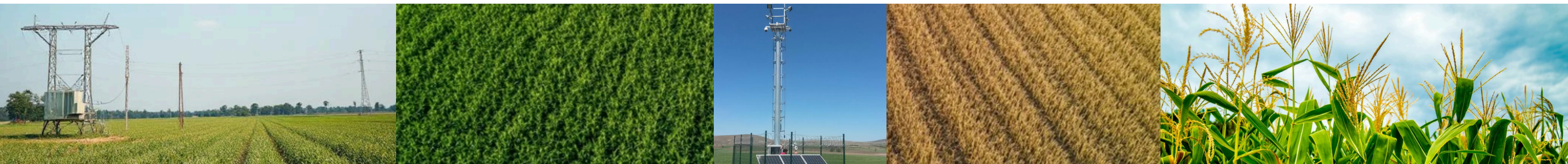
Group: FARMERS

Zeynep Kara 504251529

Mehmet Şahin 504251512

Oğuzhan Aybar 504251528

Mungunsarnai Tumurkhoo 504241572



PHENOLOGICAL TARGET SCOPE

THE PROBLEM

- Lack of Wheat farming automatization
- Phenological stage drives every farm decision (irrigation, fertilization, pesticide, harvest timing)
- Manual labeling is slow, subjective, observer-dependent
- Does not scale to country-wide station networks like TARBIL

8 Target Classes



THE GOAL

- Automatically classify 8 phenological stages (PS0–PS7) from agricultural station imagery
- Benchmark classical deep learning models and hybrid architectures

PILOT SCOPE

- 20 stations × years, ~6,900 labeled images

WHY IT MATTERS

- Yield, food security, climate adaptation

STUDY WORKFLOW & WORKSPACE

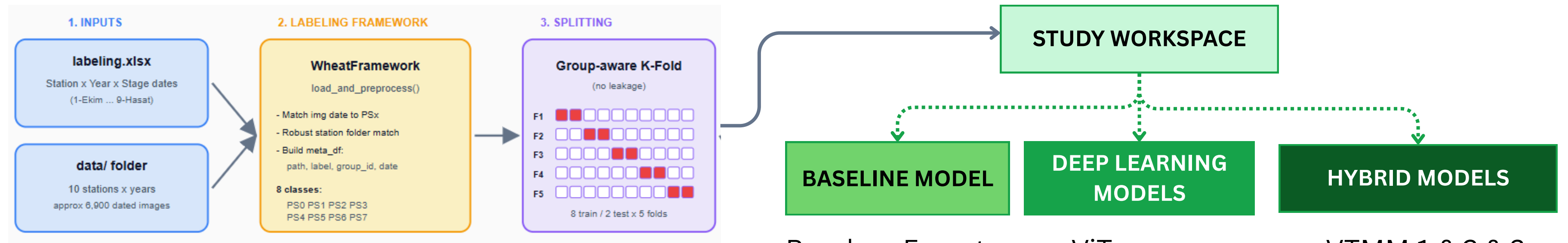


Image Collection

Agricultural station images from multiple years (20 stations)

Data Labeling

8 phenological stages (PS0–PS7)

Dataset Splitting

Group-Aware K-Fold Validation

Preprocessing & Augmentation

Resize, tiling

Evaluation

Mean Accuracy

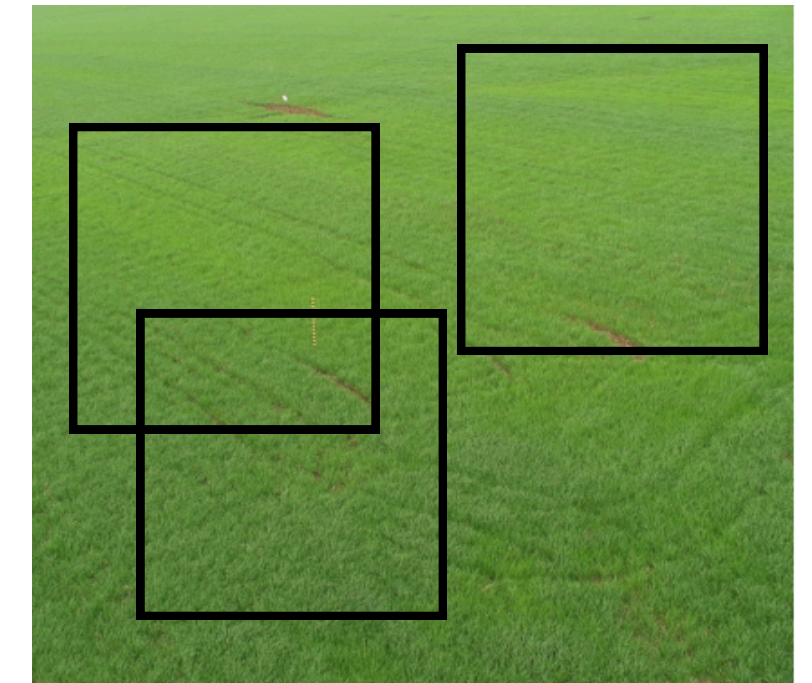
PREPROCESS METHODS

Data Structuring & Augmentation

- Resizing (224 x 224) (448 x 448)
- 4-tile partitioning
- Random single-tile selection
- Mirroring



Fixed 4-tile Partitioning



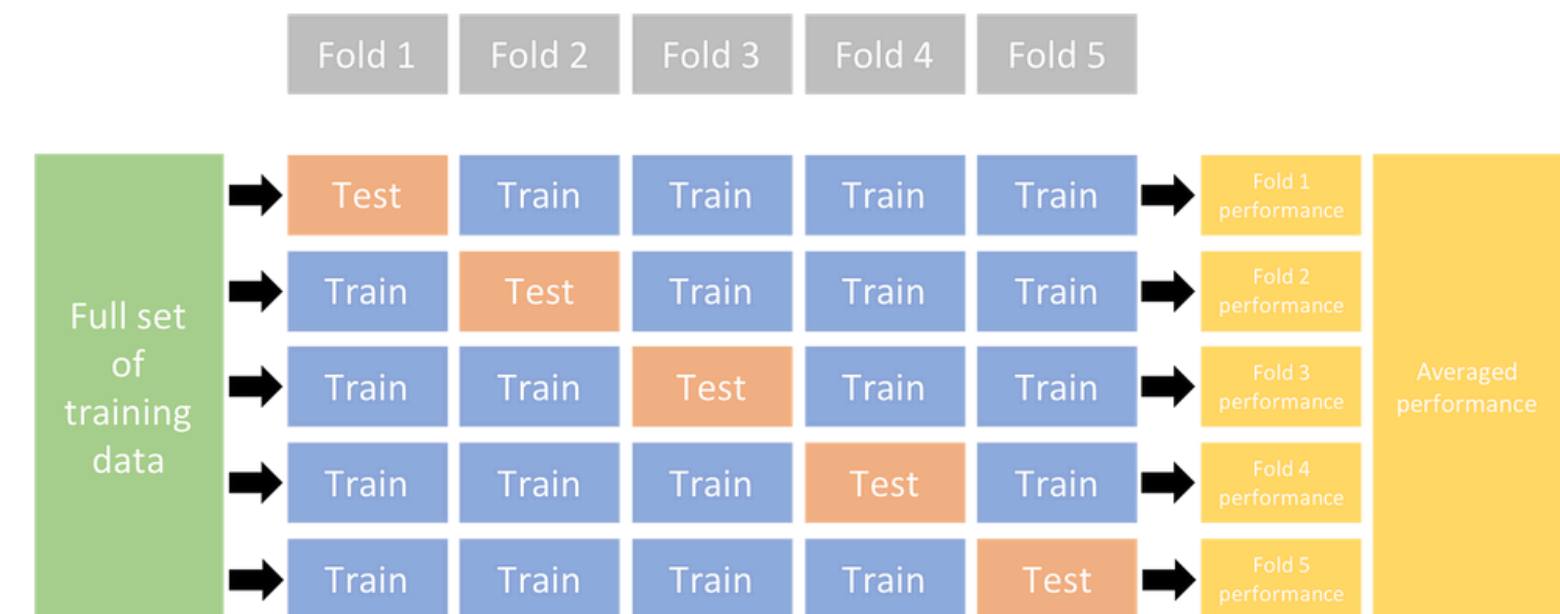
Random-tile selection

Group K-Fold Cross-Validation:

Stations are kept strictly separate across folds.

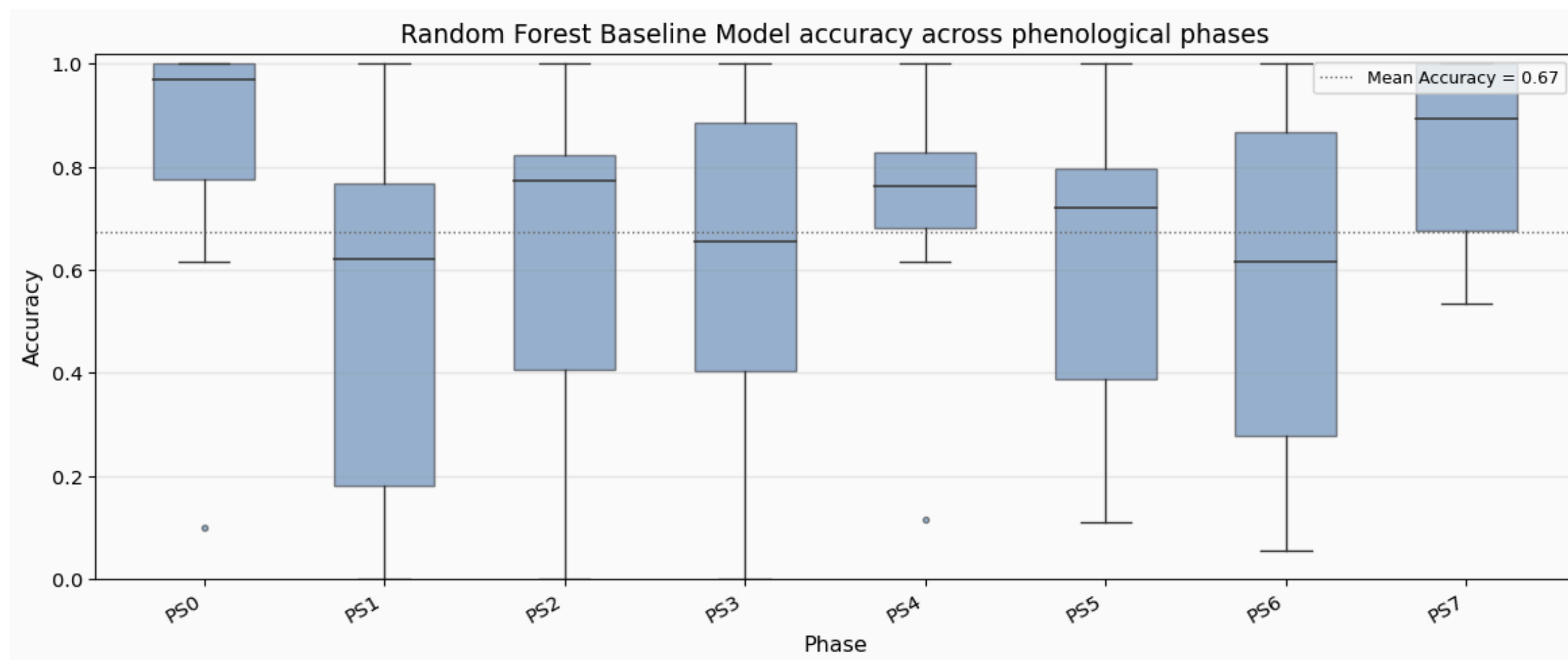
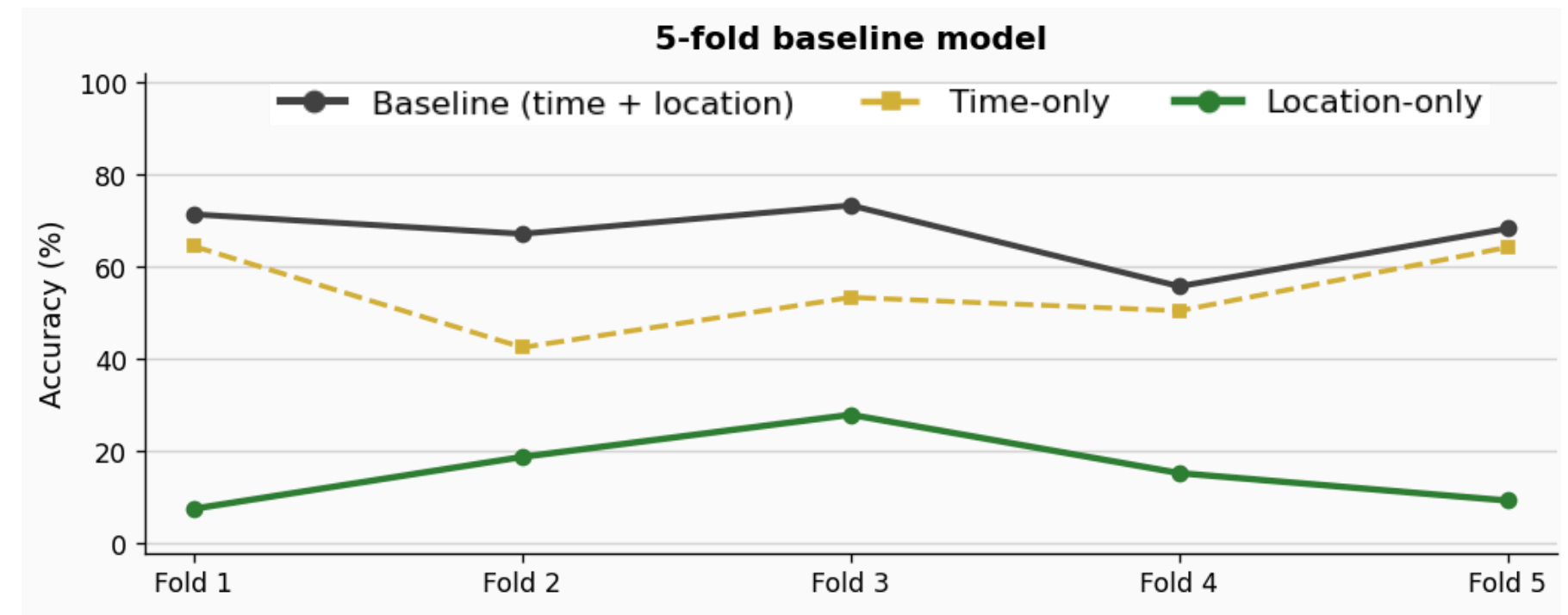
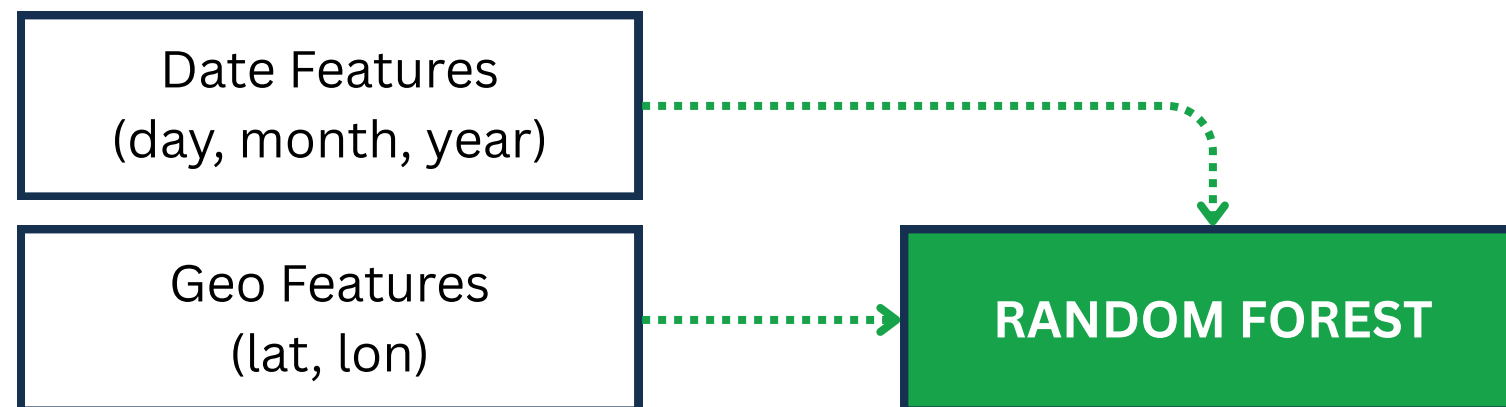
Leakage Prevention:

Photos from the same station cannot exist in both training and test sets simultaneously.



THE BASELINE MODEL

RANDOM FOREST



Baseline Model
Accuracy: 67.4%

**Baseline Model
(only date)**
Accuracy: 50%

**Baseline Model
(only location)**
Accuracy: 20%

| THE VISUAL DEEP LEARNING MODELS

The Aim:

To classify the phenological stage by processing wheat field images

	TRIED BY	COST (TIME)	MEAN ACCURACY
ViT	OĞUZHAN	~180m	%66
CNN	ZEYNEP	~190m	%54
RCNN	MEHMET	~400m	%52
YOLO	MUNGUNSARNAI	~250m	%50

HARDWARE: i9 12900KF - RTX4090

The Result:

Models did not show any improvement over baseline while increasing computation cost substantially

The Takeaway:

- Both metadata and image data independently contain interpretable indicators of PS.
- Combining them can improve performance.

Next Step:

- Select the most successful model: ViT
- Hybrid Model :
Base Model + Deep Learning Model

| THE HYBRID MODEL (VTMM)

- ViT and the metadata model act as a feature extractor
- The decider classifier combines the extracted features and produces a final result

PROS:

- Both metadata and image data

CONS:

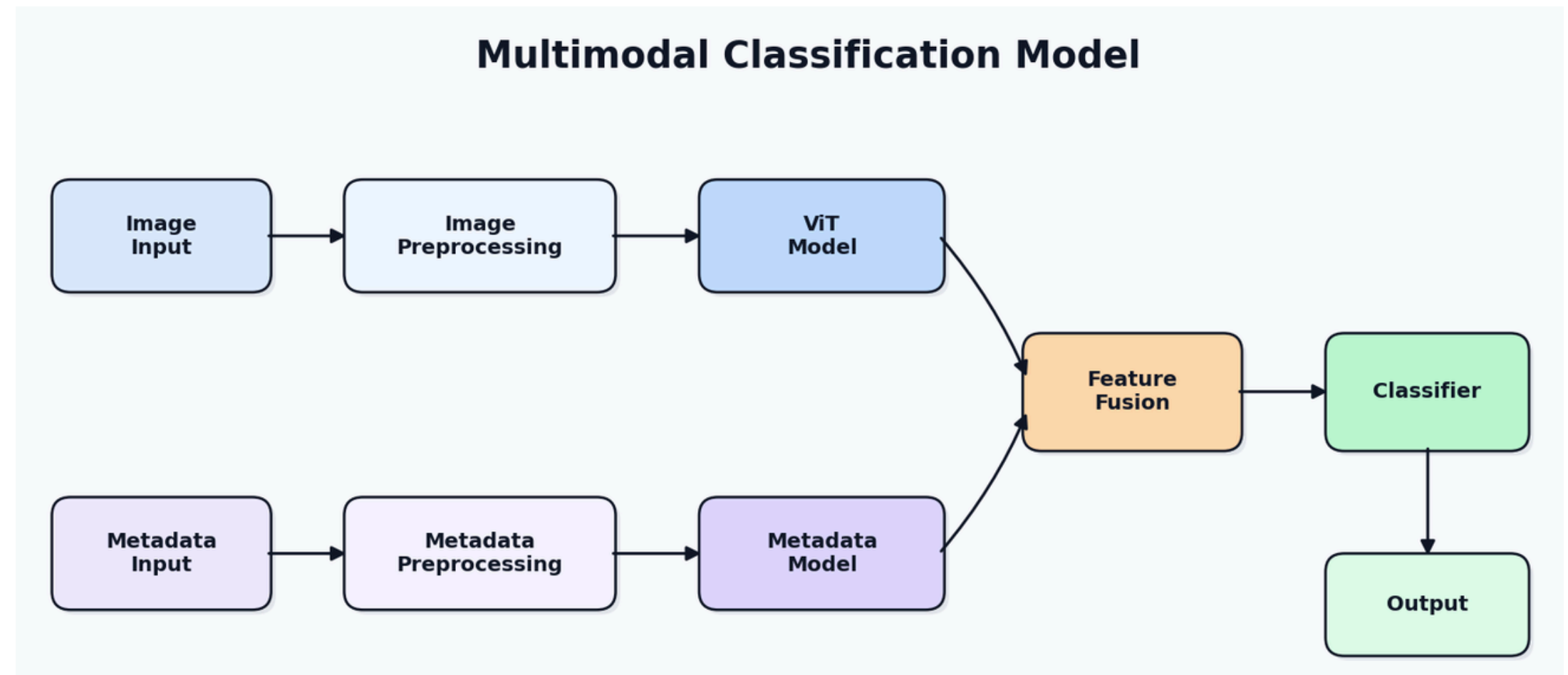
Bigger model

↓

more parameters

↓

more hungry



THE HYBRID MODEL (VTMM)

EXPECTATION: The hybrid model works better than the singular models

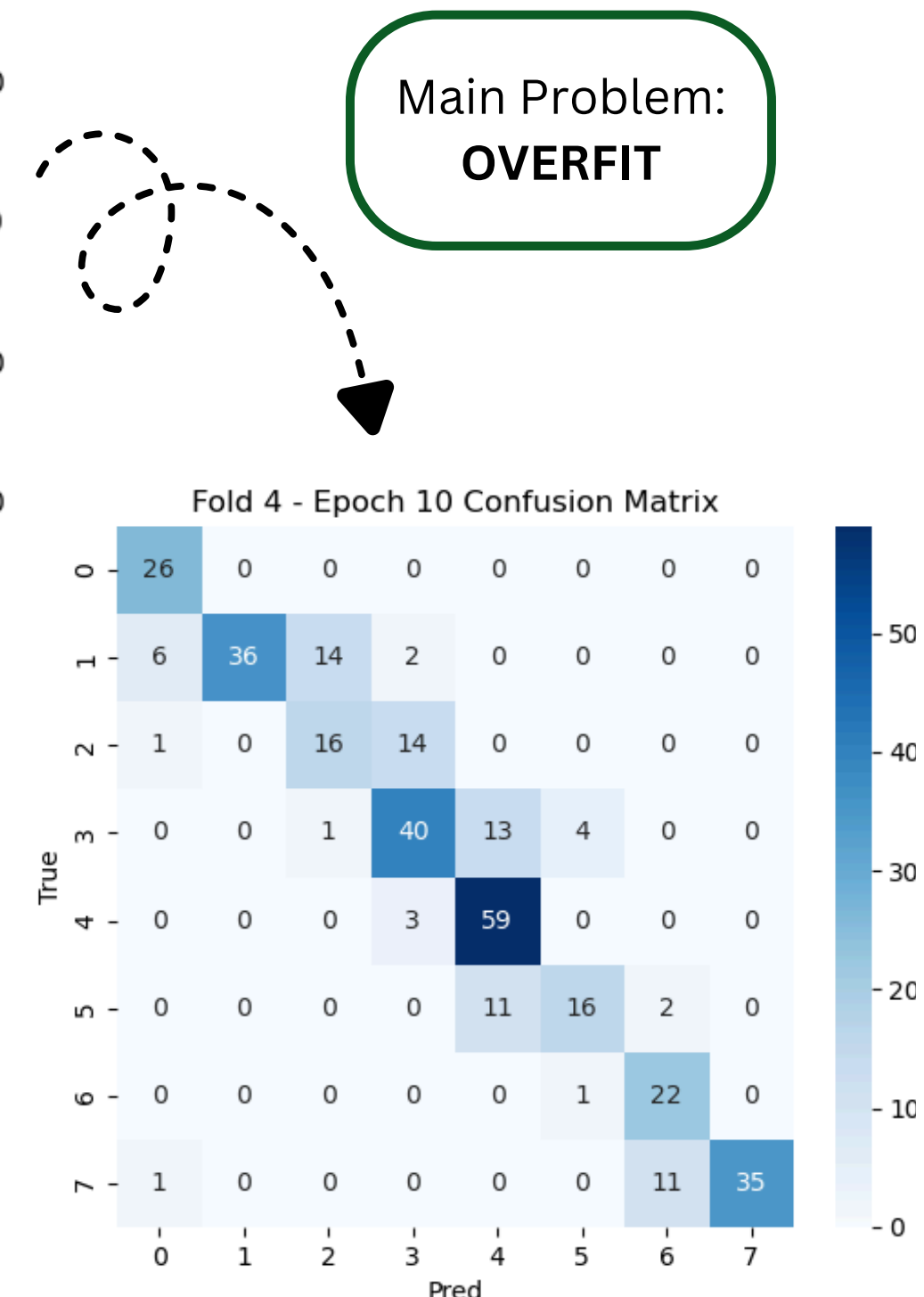
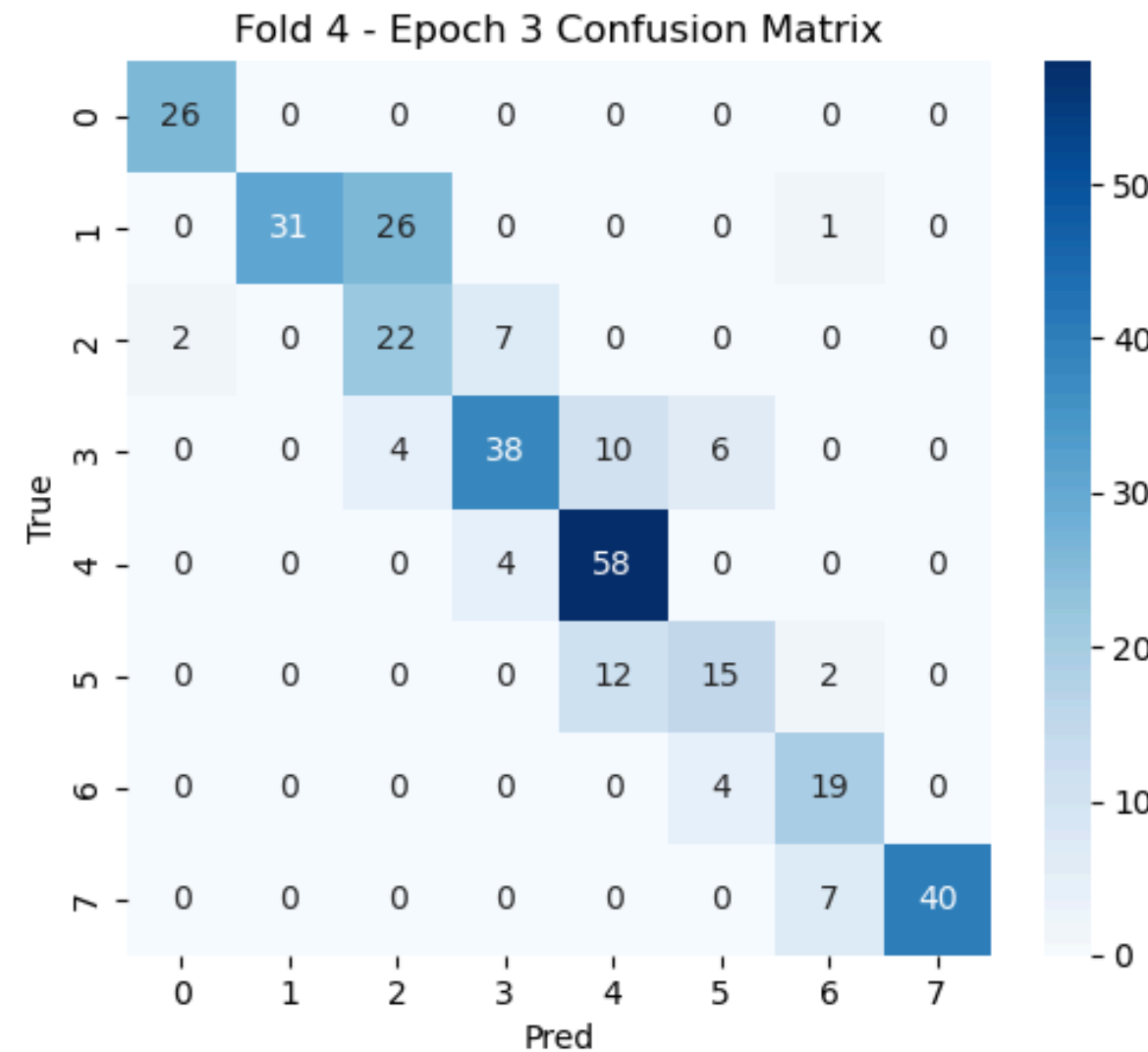
Baseline Model
Accuracy: 67.4%

ViT Model (Image)
Accuracy: 66.3%

Hybrid Model
Accuracy: 68.1%

TAKEAWAY: No significant improvements may mean:

- Overfitting
- Not enough data to generalize
- The data is not representative of the target



THE HYBRID MODEL VARIANTS

Accuracy:
60 - 66%

The best accuracy:
85% (not reliable)

VTMM V1

TRIED TO SOLVE:

- no metadata usage alongside image

BY:

- combining metadata and image data with a decider layer

RESULT:

- no accuracy improvements

VTMM V2

TRIED TO SOLVE:

- overfitting

BY:

- hyper-parameter tuning and model freezing

RESULT:

- no improvements
- less overfitting

VTMM V3

TRIED TO SOLVE:

- overfitting

BY:

- by randomly selecting sub-images from the source image (random tiling)

RESULT:

- no improvements
- reduced overfitting

VTMM V4

TRIED TO SOLVE:

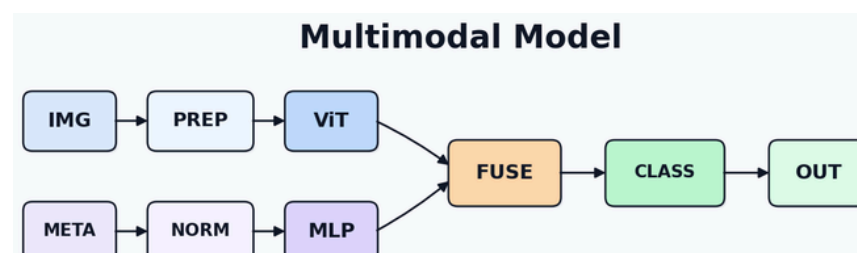
- not being able to extract valuable information from the images

BY:

- Utilizing multiple ViT to create a crude MoE

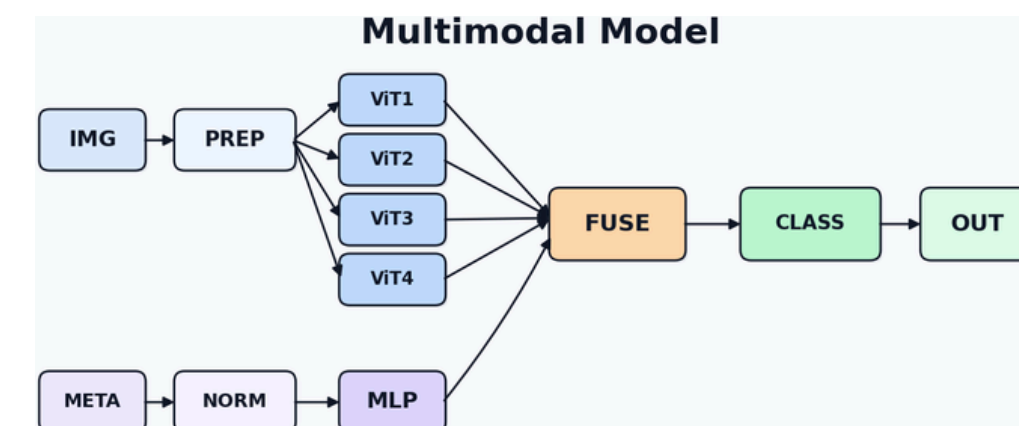
RESULT:

- no improvements



VTMM V2 =
VTMM 1 + finetuning

VTMM V3 =
VTMM 2 +
(random tiling)



THE HYBRID MODEL V5.1

WHY PREDICT FOR EVERY IMAGE WHEN ONLY STAGE-CHANGE DATES MATTER?

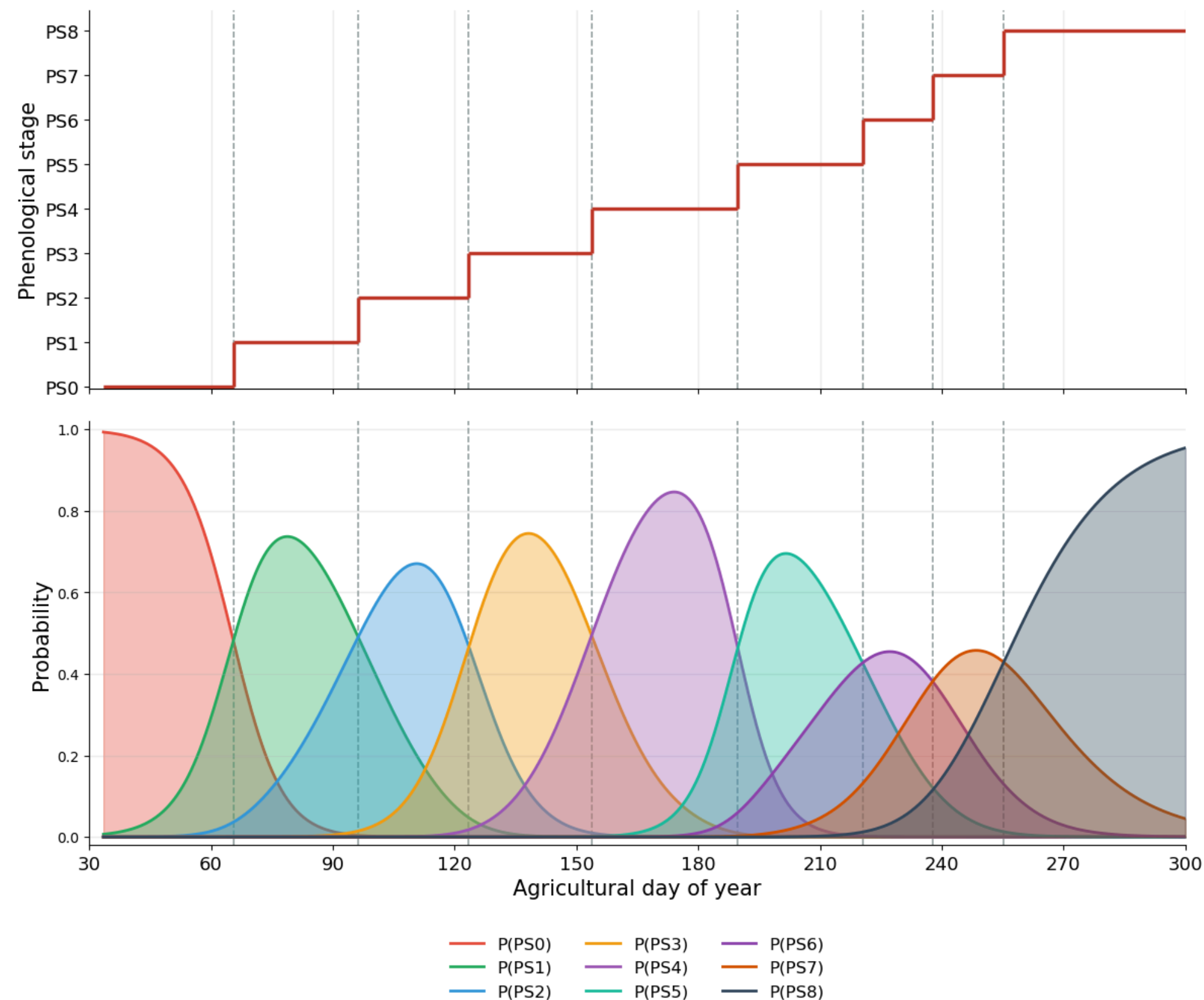
The Idea: A **whole year's** worth of data can be represented with **9 dates** (phenological stage starting dates), so focus on these 9 dates.

The Aim: Instead of $\text{argmax}(P(\text{PS}(t)))$,
 $P(\text{PS}(t+1)) > P(\text{PS}(t))$

The Result: The errors **disastrously** propagated and accumulated

Accuracy:
56.72%

✨ **Food for Thought** ✨ : Deciding on the definite state change is disastrous, but a **probabilistic agri DOY-based weighting** with **rolling average** may have worked a lot better.



THE HYBRID MODEL V5.2

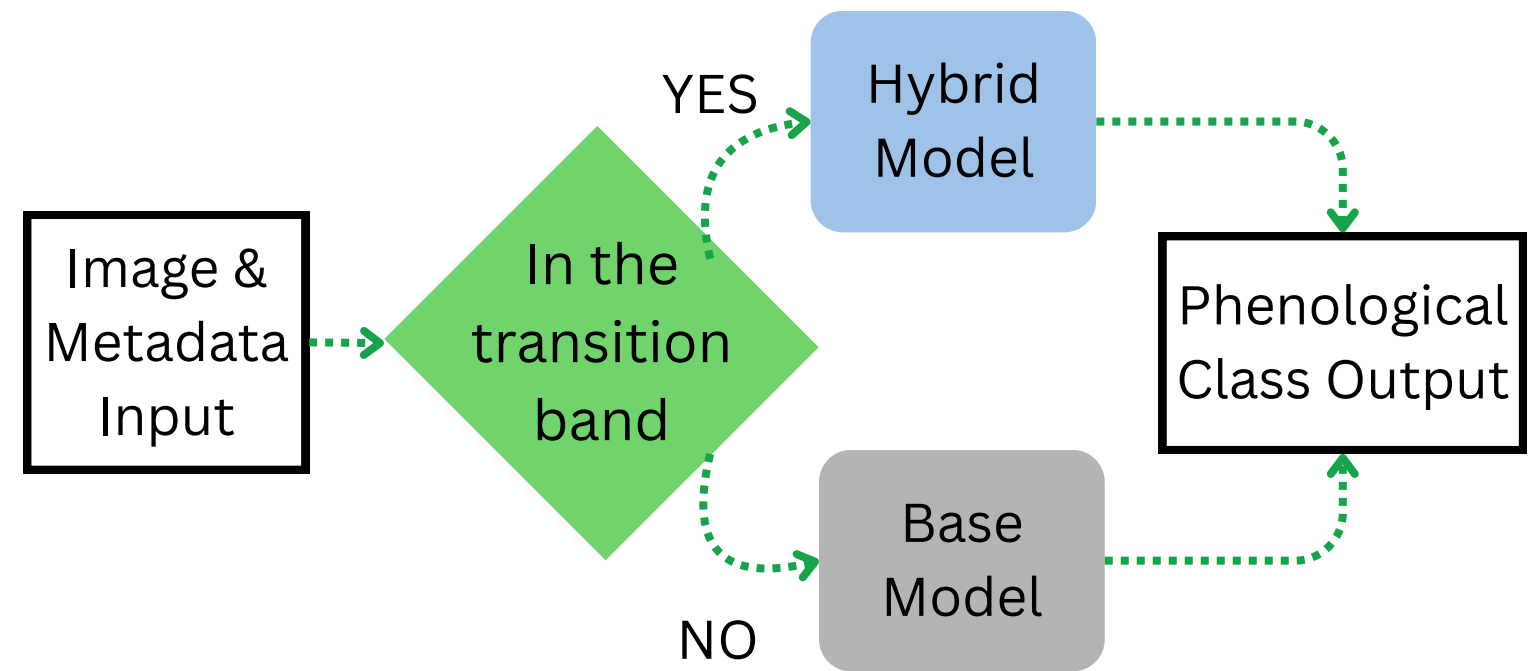
GATED VTTM + BASELINE

Baseline Model
Transition Accuracy: **49.62%**

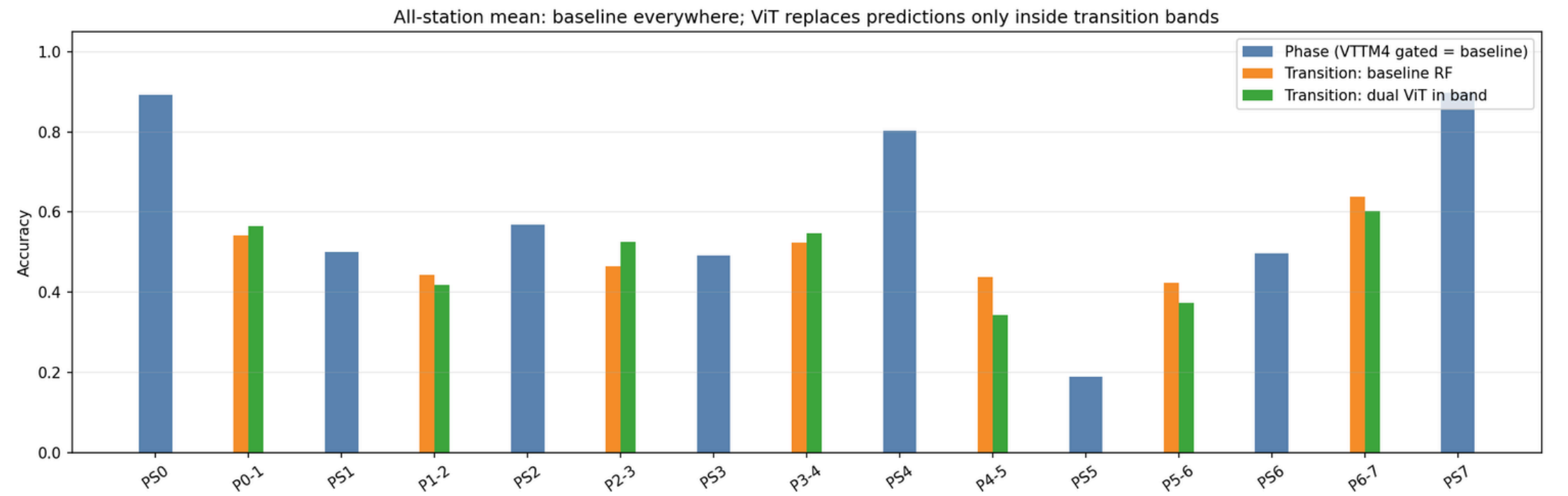
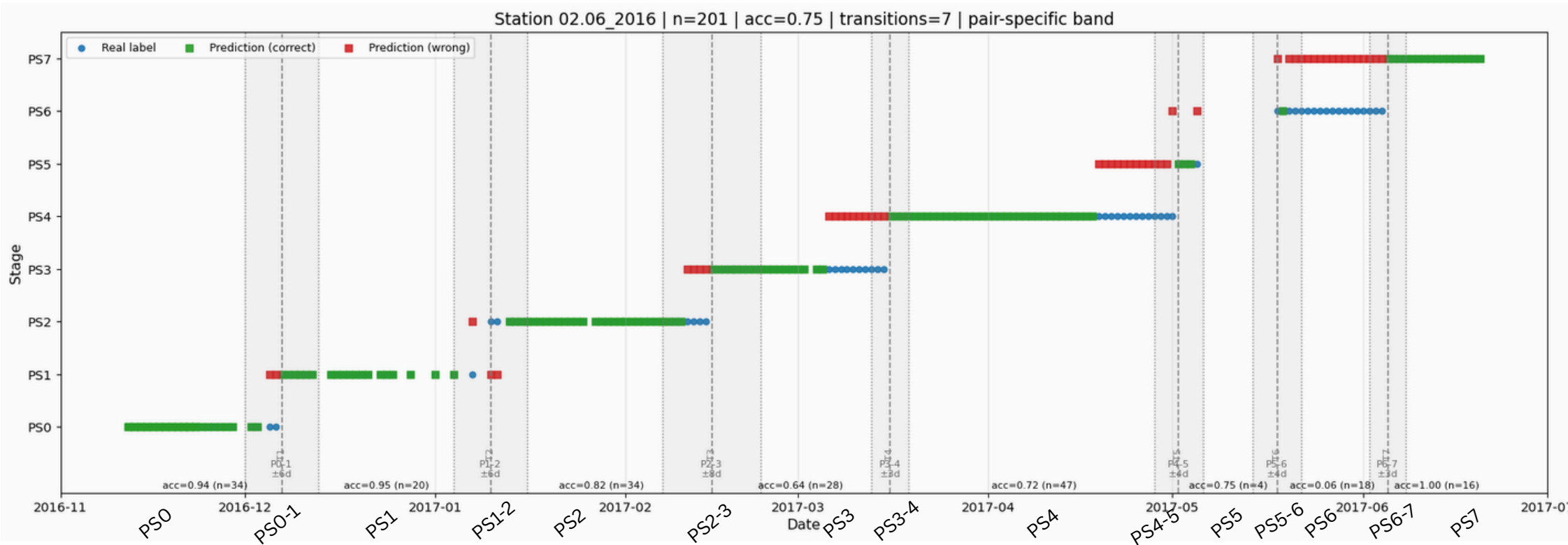
Hybrid Model
Transition Accuracy: **48.22%**

The Aim: Strategic Gating

Leveraging the baseline's macro-trend reliability while deploying the hybrid model to resolve transition-band uncertainty.



Average Transition Phase Bands
 'P0-1': 6 'P1-2': 6, 'P2-3': 8
 'P3-4': 4 'P4-5': 4 'P6-7': 3



THE HYBRID MODEL V6

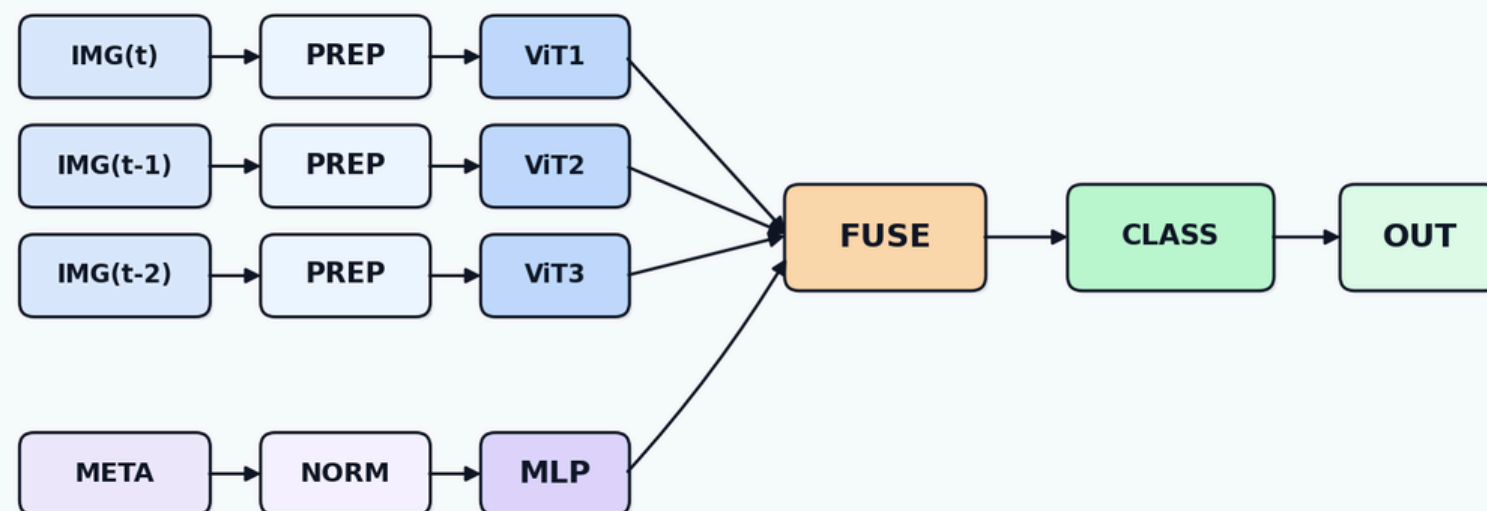
THE IMAGES TREATED AS TIME-SERIES DATA

The Idea: Images are a **time-series data** like a video, not independent samples.

The Aim: Use the last 3 days' image to clearly see recent changes.

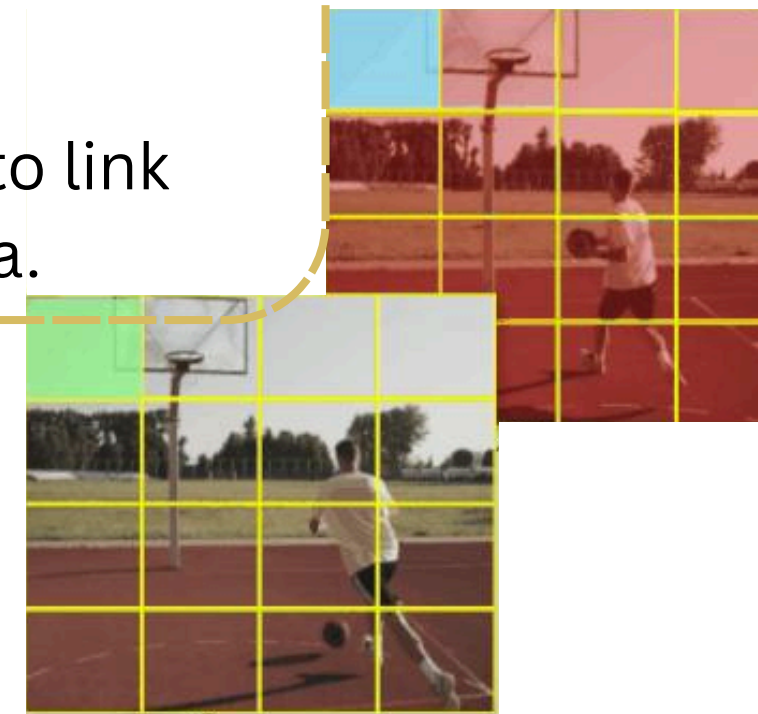
Accuracy:
64.72%

Multimodal Model

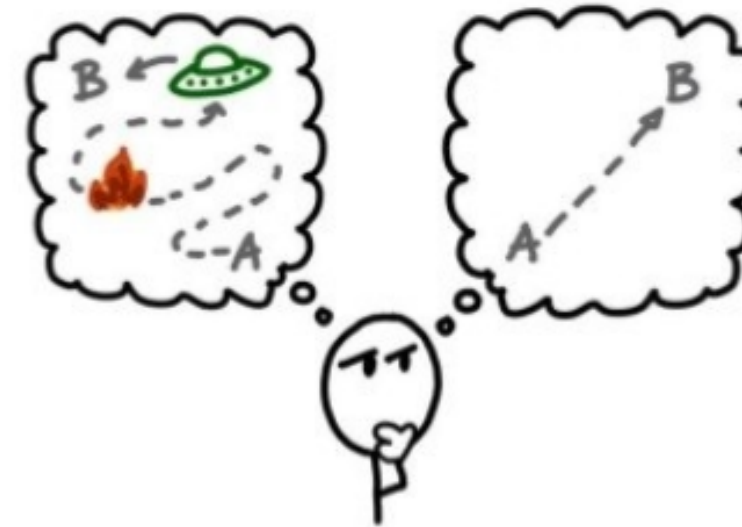


What the Future Holds

- **Untested Idea:** If the images form a time series, we can model temporal relationships between images using attention across frames.
- **Goal:** Leveraging frame-to-frame attention (not only pixel to pixel) to infer phenological changes across time.
- **Video Transformers:** Leveraging architectures designed for video to link temporal context with spatial data.



THE FINAL TAKEAWAYS



THE BEST MODEL
Baseline Model
Accuracy: 67.4%

The Challenges of Visual Data in Phenology

- **Station Heterogeneity:** Non-standard camera angles and hardware across sites
- **Small Sample Constraints:** High variance leads to overfitting in complex models.
- **Environmental Noise:** Inconsistent lighting and weather obscuring plant features.

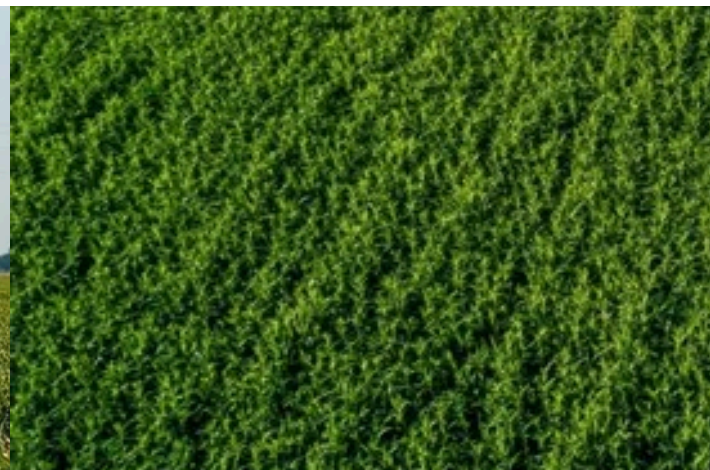
Why "Simple" Often Wins: Occam's Razor

- **Metadata as a Climatic Proxy:** Date and Location encapsulate the climatic DNA
- **Diminishing Returns:** Metadata already provides the "macro" answer
- **Robustness:** Images introduce "noisy" pixels

BLGE 521 AI

Group: FARMERS

THANK YOU FOR LISTENNING



Wheat phenology stage

Field photo → predicted growth stage. **Date** is optional if the filename contains `YYYY-MM-DD`, `YYYY_MM_DD`, or `YYYYMMDD` (same as training paths). **Coordinates** are optional if the name starts with a Turkish plate code (`06_...`, `34_...`); otherwise defaults match the notebook (province 6 centroid).

Model ready · 8 classes · center · cuda · uploads ≤ 200 MB

Image

Dosya Seç Dosya seçilmedi

Date OPTIONAL IF IN FILENAME

gg . aa . yyyy



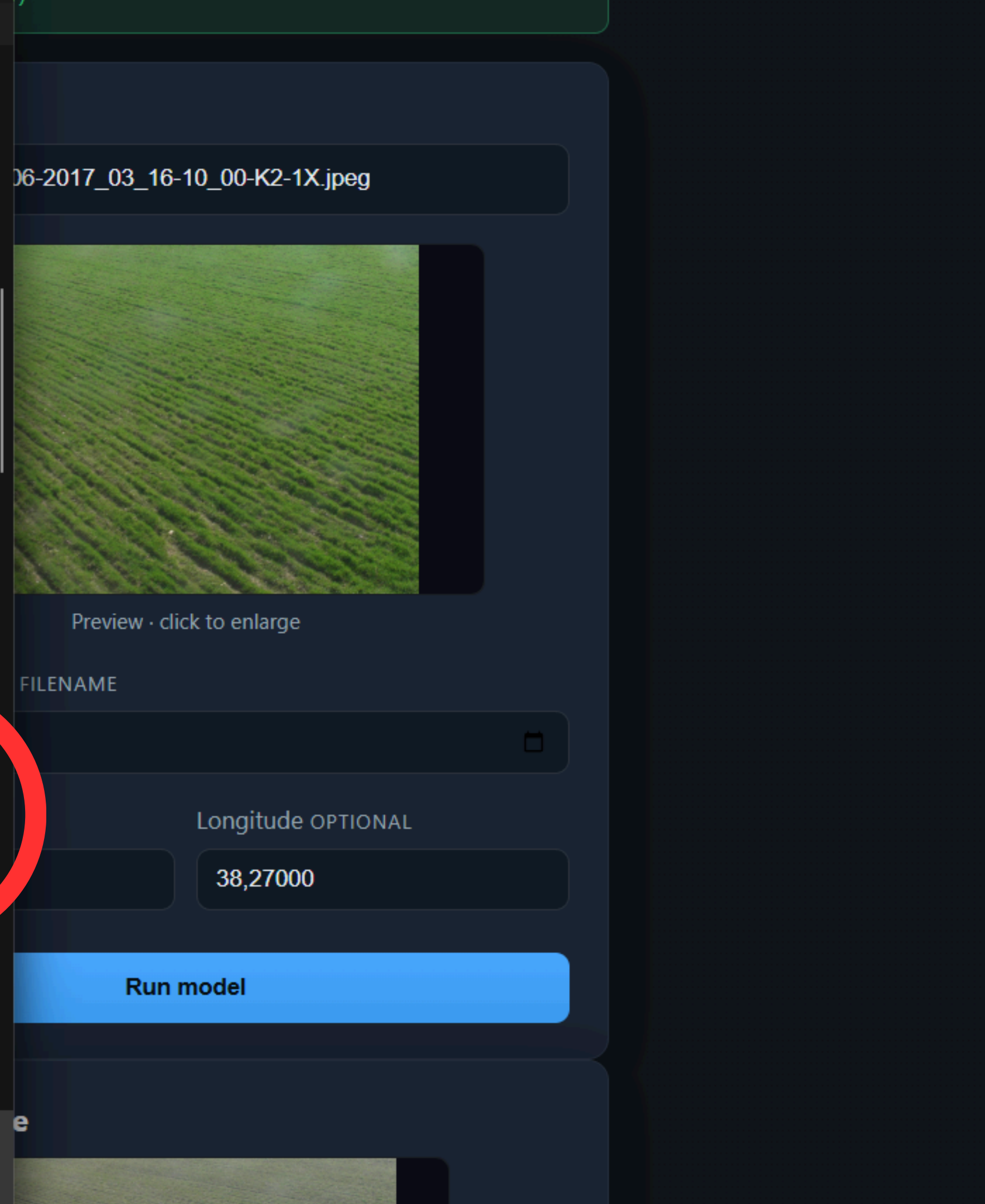
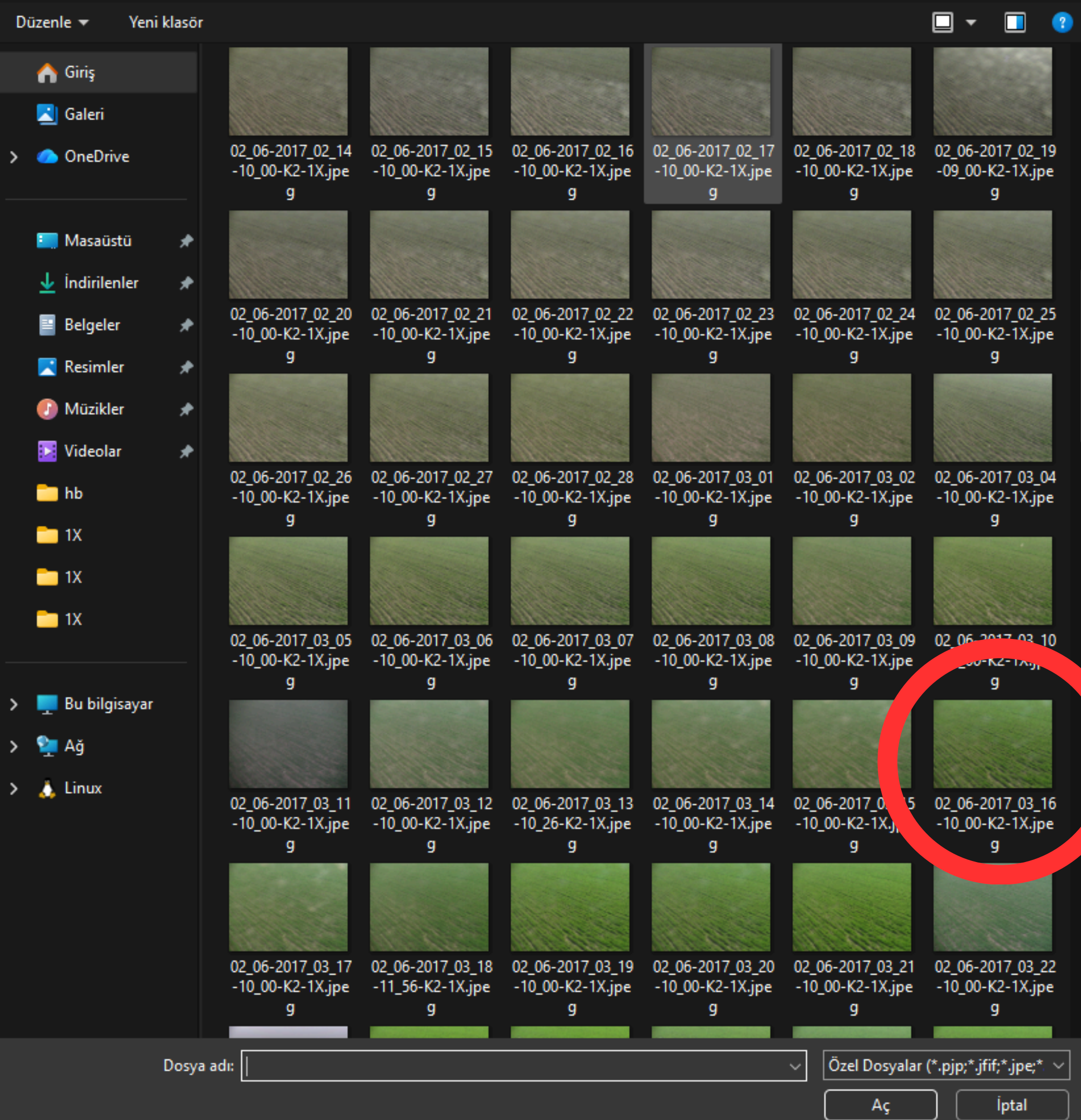
Latitude OPTIONAL

Longitude OPTIONAL

e.g. 39.93 · leave blank to use

e.g. 32.85

Run model



VIEW MODEL

Predicted stage



Click to enlarge

PS4

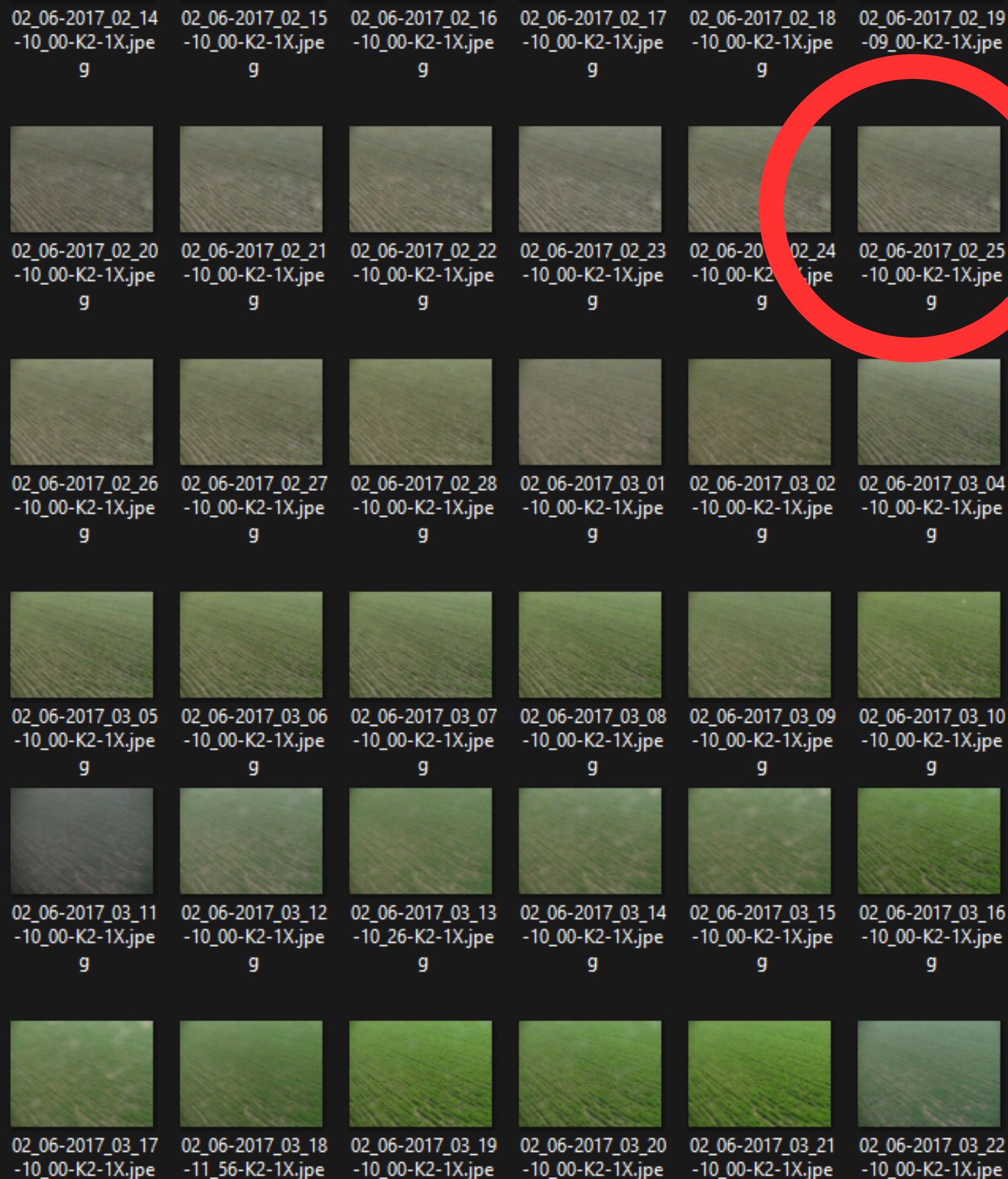
Confidence 76.4% · date 2017-03-16 · manual · agri DOY 136 ·
37.7600, 38.2700 · manual · center

Top probabilities

- PS4: 76.4%
- PS3: 17.6%
- PS6: 1.5%
- PS2: 1.4%
- PS5: 1.3%
- PS0: 0.8%
- PS1: 0.6%
- PS7: 0.4%



Yeni klasör



Dosya adı:

Özel Dosyalar (*.pjp;*.jfif;*.jpe;*)

Aç

İptal

ready (center)

Seç 02_06-2017_02_25-10_00-K2-1X.jpeg



Preview · click to enlarge

OPTIONAL IF IN FILENAME

02.2017

Latitude OPTIONAL

5000

Longitude OPTIONAL

38,27000

Run model

Run model

Predicted stage



Click to enlarge

PS2

Confidence 59.9% · date 2017-02-25 · manual · agri DOY 117 ·
37.7600, 38.2700 · manual · center

Top probabilities

- PS2: 59.9%
- PS3: 29.3%
- PS4: 2.9%
- PS6: 2.5%
- PS1: 2.4%
- PS0: 1.2%
- PS7: 1.1%
- PS5: 0.7%



Station Code	Year	1-Ekim	2 - Çıkış	3 - Çimlenme	4 - Kardeşlenme	5 - Sapa Kalkma	6 - Başaklanma	7 - Çiçeklenme	8 - Olgunlaşma	9 - Hasat
2.06	2016	12.11.2016	7.12.2016	10.01.2017	15.02.2017	16.03.2017	2.05.2017	16.05.2017	5.06.2017	20.06.2017



First Image
16.03.2017
PS4



Second Image
25.02.2017
PS4